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ORIGINAL ARTICLE

In silico prediction of drug-induced cardiotoxicity with ensemble machine learning and structural pattern recognition

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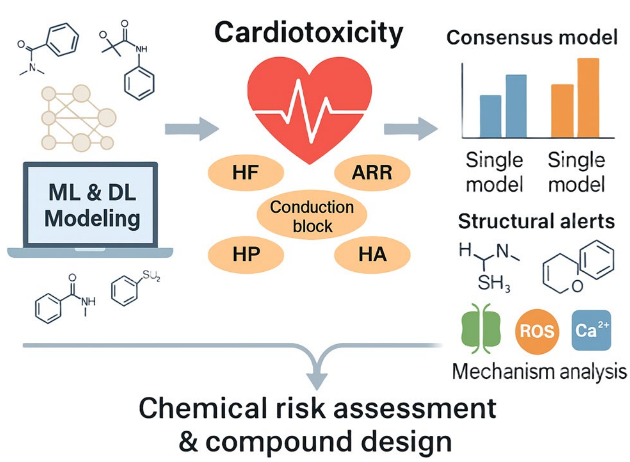
Abstract
Drug-induced cardiotoxicity poses a significant risk to human health, and reliable predictive models are needed for safety assessment. In this study, a range of machine and deep learning models were developed for five cardiotoxicity end points, including heart failure (HF), arrhythmia (ARR), heart block (HB), hypertension (HP), and heart attack (HA). A total of 110 predictive models were constructed for each cardiotoxicity endpoint using various algorithms and molecular descriptors, and consensus models were developed based on the best-performing individual classifiers. The consensus models consistently outperformed individual models in cross-validation and external validation. Further molecular property analysis revealed that cardiotoxic compounds tend to exhibit higher molecular weight, increased lipophilicity (logP), lower hydrogen bonding capacity (HBD and HBA), and reduced topological polar surface area (TPSA). Additionally, key structural alerts (SAs), including secondary amines, benzene derivatives, sulfonamide/sulfonylurea groups, and heterocyclic structures, were identified. These SAs may mediate cardiotoxicity through ion channel inhibition, oxidative stress induction, and calcium homeostasis disruption. This study provides an integrated machine learning and deep learning computational framework for drug cardiotoxicity assessment and provides an exploration of the structural characteristics of cardiotoxic compounds, which is helpful for the discovery of safer drugs and chemical risk assessment.

Graphical abstract



Keywords Drug-induced cardiotoxicity · Artificial intelligence model · Toxicity mechanisms · Structural alerts · Chemical risk assessment

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Introduction

Drug-induced cardiotoxicity is a significant challenge in the risk assessment of pharmaceuticals, industrial chemicals, and the environment chemicals, potentially leading to severe health complications such as arrhythmias, heart failure, and sudden cardiac death. The cardiotoxicity assessment methods primarily always rely on preclinical animal studies and clinical trials. However, these methods are not only expensive and time-consuming but also have limitations in accurately reflecting human responses. To address these shortcomings, increasing efforts have been directed toward computational toxicology methods [1–4], particularly the application of machine learning (ML) and deep learning (DL) techniques in toxicity prediction.

Over the past few decades, machine learning has made remarkable progress for the safety assessment in drug discovery. In particular, the quantitative structure–Activity relationship (QSAR) model has been widely applied in structure-based prediction of drug toxicity. For instance, researchers can develop computational models based on molecular descriptors to predict whether a compound has hERG channel inhibitory toxicity [5–7]. Many studies have screened compound databases to identify chemical structures that may be related to cardiotoxicity and have verified the QSAR model using experimental data [8]. This approach may have limitations when handling highly complex relationships between compounds and toxicity. In 2018, Cai et al. [9] reported on a novel computational model designed to systematically identify drug-induced cardiovascular complications, including five common end points: hypertension, heart block, arrhythmia, heart failure, and myocardial infarction. This study employed five machine learning algorithms (logistic regression, random forest, K-nearest neighbor, support vector machine, and neural network) to develop an integrated classifier. The performance of this integrated classifier was superior to that of a single model, with AUC values ranging from 0.784 to 0.842. In 2022, Iftkhar et al. [2] developed CardioToxCSM, a graph convolutional network (GCN)-based method for the prediction of multiple cardiotoxic outcomes. The model had high efficiency and accuracy. The AUC was 0.79 by tenfold cross-validation, which was consistent with the blind test results. This approach is outstanding in learning molecular structural information, identifying potential cardiotoxicity, and generating reliable predictions to help select drugs with better cardiac safety. In recent years, perturbation theory machine learning (PTML) models, especially the subclasses known as multitasking QSAR, have significantly advanced the field of simultaneous prediction of pharmacological efficacy and toxicity profiles. PTML models offer the advantage of integrating multiple bioactivities or safety endpoints into a unified and interpretable modeling framework. For instance, PTML-based multitasking QSAR models have been successfully applied to predict antibacterial potency alongside ADMET properties, model acute toxicity, and support de novo compound design with favorable safety profiles [10–16]. These models not only allow simultaneous safety profiling but also provide interpretable insights by identifying molecular fragments associated with both efficacy and toxicity modulation. Integrating the principles of PTML may help accelerate early-stage drug discovery by supporting virtual screening and rational molecular optimization. In our previous study [17], we developed an interpretable deep neural network model to predict chemical cardiotoxicity associated with the inhibition of hERG, Nav1.5, and Cav1.2 channels. This model combined large-scale chemical and experimental data to accurately predict the inhibitory effects of compounds on different cardiac ion channels. Compared with traditional methods, machine learning and deep learning models demonstrate greater flexibility and accuracy in dealing with complex molecular structures and predicting multi-dimensional toxic outcomes [18, 19].

Despite significant progress made in the prediction of drug-induced cardiotoxicity, there are still some challenges. Most existing models are difficult to comprehensively predict various types of cardiotoxicities. These models often focus on a single type of toxicity, which limits their applicability. Furthermore, although interpretability is becoming increasingly important in deep learning models, many models still suffer from the "black box" problem, which makes it difficult to provide clear mechanism explanations. Structural feature recognition remains a key issue in the prediction of cardiotoxicity and requires further research.

The present study aims to develop machine learning and deep learning models for the prediction of drug-induced cardiotoxicity, with a particular emphasis on analyzing structural features related to cardiotoxic effects. We not only developed the predictive models but also systematically identified the structural alerts related to cardiotoxicity. These structural insights will help clarify the potential toxicity mechanisms and provide valuable assistance for the evaluation of the cardiotoxicity in drug discovery and development.

Materials and methods

Data collection and preparation

This study focuses on five key cardiotoxicity endpoints, including heart failure (HF), arrhythmia (ARR), heart block (HB), hypertension (HP), and heart attack (HA). The dataset used for model construction and structural feature analysis was obtained from the studies conducted by Cai et al. [9] and Saba Iftkhar et al. [2], which compiled cardiotoxicity-related compounds from multiple publicly available databases, including the Comparative Toxicogenomics Database (CTD), Side Effect Resource (SIDER), Off-Label Side Effect (OFFSIDES), and MetaADEDB.

To standardize the dataset, several preprocessing steps were used: (1) identification and removal of redundant chemical entries across multiple databases based on InChIKey identifiers; (2) RDKit [20] was used to optimize the chemical structure, including desalting, charge neutralization, and tautvariant standardization; (3) for compounds with multiple structural representations, only biologically relevant active metabolites are retained to enhance their physiological relevance; (4) to address the potential class imbalance between cardiotoxic and non-cardiotoxic compounds, a stratified sampling strategy was adopted to maintain a balanced distribution between the training and validation sets. All chemical structures were stored in SMILES format.

Model development

The model development in this study was conducted using the Online Chemical Modeling Environment (OCHEM) platform [21, 22]. OCHEM is a web-based database and modeling platform designed to automate and streamline the essential steps required for QSAR modeling. The platform integrates various widely used or state-of-the-art machine learning and deep learning algorithms, enabling the development of self-learning computational models that can access and analyze data for predictive purposes.

For machine learning modeling in this study, seven widely applied algorithms were selected, including Associative Neural Network (ASNN), Random Forest Regression and Classification (RFR), Least Squares Support Vector Machine (LSSVM), Fused Sparse Multinomial Logistic Regression (FSMLR), J48 Decision Tree Algorithm (WEKA-J48), k-Nearest Neighbors (kNN), and Extreme Gradient Boosting (XGBoost). These algorithms have been extensively applied in computational toxicology and QSAR modeling due to their robustness and ability to handle high-dimensional chemical descriptors [23–27]. Deep learning, as an extension of artificial neural networks, demonstrates superior feature learning capabilities. DL algorithms can automatically extract hierarchical representations from molecular data, learn statistical patterns from large training samples, and improve predictive accuracy on unseen data. In this study, five DL methods available on the OCHEM platform were employed, including TEXTCNN (from DeepChem), Transformer Convolutional Neural Network (TRANSNN), CNF, KGCNN AttFP, and KGCNN ChemProp. These models have shown promising performance in predicting molecular properties and toxicological endpoints [28, 29].

To generate molecular descriptors for ML and DL models, OCHEM provides access to various descriptor packages. In this study, 15 different descriptor sets were selected, including ALogPS, Oestate, CDK23, Dragon6 (2D blocks: 1–28), Dragon7 (3D blocks: 1–30), EPA (T.E.S.T.), Fragmentor (length: 2–4), InductiveDescriptors, JPlogP, MOLD2, MORDRED, QNPR (length: 1–3), RDKIT (3D blocks: 1–11, 15–16), RDKIT (ECFP4), Structural Alerts, and alvaDesc (3D blocks: 1–33). These descriptors encompass a wide range of physicochemical, topological, and structural properties that are important for toxicity prediction.

Based on these features, a total of 110 individual models were developed for each cardiotoxicity endpoint, including 105 machine learning models (combinations of seven ML algorithms with 15 molecular descriptor sets) and five deep learning models. The best-performing individual models were selected based on cross-validation results and further integrated into a consensus model to enhance predictive performance. The consensus modeling strategy has been widely used in computational toxicology, as it combines the strengths of multiple individual models, reduces prediction variance, and improves generalization to external datasets [30, 31].

All models were optimized using the default parameter settings provided by OCHEM platform.

Model evaluation

The prediction performance of the model was evaluated with fivefold cross-validation and an independent external validation set. In the fivefold cross-validation, the dataset is randomly divided into 5 subsets. In each iteration, 4 subsets are used for training and the other 1 subset is used for testing. This process is repeated five times to ensure that each subset is used at least once as a test set and to record the average performance of all iterations. The external validation set (composed of compounds completely independent of the model training process) is used to evaluate the generalization ability of the model and ensure its reliability in applications.

To comprehensively evaluate model performance, several statistical metrics were employed (Eqs. 1–5).

$$\mathrm{ACC} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FN} + \mathrm{TN} + \mathrm{FP}}, \tag{1}$$

$$\mathrm{SE} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}, \tag{2}$$

$$\mathrm{SP} = \frac{\mathrm{TN}}{\mathrm{TN} + \mathrm{FP}}, \tag{3}$$

$$\mathrm{BAC} = \frac{\mathrm{SE} + \mathrm{SP}}{2}, \tag{4}$$

$$\mathrm{MCC} = \frac{\mathrm{TP} * \mathrm{TN} - \mathrm{FP} * \mathrm{FN}}{\sqrt{(\mathrm{TP} + \mathrm{FP})(TP + \mathrm{FN})(\mathrm{TN} + \mathrm{FP})(\mathrm{TN} + \mathrm{FN})}}, \tag{5}$$

where TP denotes true positives, FP denotes false positives, TN denotes true negatives, and FN denotes false negatives.

Accuracy (ACC) was used to measure overall classification correctness. Sensitivity (SE, recall) evaluated the model's ability to correctly identify cardiotoxic compounds, while specificity (SP) assessed the correct classification of non-cardiotoxic compounds. Balanced accuracy (BA) was calculated as the average of sensitivity and specificity to mitigate potential bias caused by class imbalance. Additionally, the Matthews Correlation Coefficient (MCC) was incorporated as a robust metric that considers true positives, false positives, true negatives, and false negatives, providing a more balanced model evaluation. Since class imbalance is common in toxicological datasets, BA and MCC were prioritized as key evaluation metrics, as they provide more reliable assessments than ACC in imbalanced scenarios. Furthermore, ROC (Receiver Operating Characteristic) curves were plotted to visualize the trade-off between sensitivity and specificity, with the area under the ROC curve (AUC) used as an indicator of overall classification performance.

For each endpoint of cardiotoxicity, we selected the model with the best performance based on its predictive performance. Using these top-level models, a consensus model is constructed by calculating the simple average of their predictions. Consensus model is an ensemble learning method, aiming to improve prediction performance and stability by combining the predictions of multiple individual models. Compared with a single model, the consensus model reduces bias, lowers variance, and improves general applicability to unseen data. Consensus models provide more reliable predictions by effectively integrating the advantages of different algorithms and are usually superior to individual models, especially in complex toxicological prediction tasks.

Structural feature analysis

A series of physicochemical descriptors were computed to characterize the molecular features of cardiotoxic compounds. The selected molecular properties included molecular weight (MW), octanol–water partition coefficient (logP), hydrogen bond donors (HBD), hydrogen bond acceptors (HBA), topological polar surface area (TPSA), and rotatable bond count (RBC). These descriptors were calculated using PaDEL-Descriptor [32], an open-source tool commonly used for molecular descriptor computation. The Wilcoxon rank sum test was used to compare whether there were intergroup differences in the molecular characteristics of cardiotoxic and non-cardiotoxic compounds. Descriptors with a p value < 0.05 were considered statistically significant.

Structural alerts (SAs) refer to molecular substructures closely associated with specific toxicity endpoints [33–35]. In this study, a frequency-based approach was employed to identify SAs related to cardiotoxicity. Specifically, predefined molecular fragments were extracted from both cardiotoxic and non-cardiotoxic compounds, and their relative occurrence was analyzed using F-score and frequency ratio (FR) methods. The Klekota-Roth fingerprint (KRFP) was used as the reference framework, which has been widely applied in toxicity prediction studies. Molecular fragments that appeared in at least six cardiotoxic compounds were retained for further analysis. Structural alerts with an F-score ≥ 0.002 and a positive rate (PR) ≥ 0.80 were identified as key contributors to cardiotoxicity.

Results and discussion

Molecular diversity analysis

This study retained 1201, 2169, 1488, 2228, and 1227 data records for the cardiotoxicity endpoints heart failure (HF), arrhythmia (ARR), heart block (HB), hypertension (HP) and heart attack (HA), respectively. All the chemical structures are given in Table S1–S5. Due to the differences in the balance of positive and negative samples in different datasets, in order to ensure the effectiveness of model training and the representativeness of the datasets, we have adopted different segmentation strategies for the training set and the validation set, as shown in Table 1. For HF data with small dataset, a 9:1 segmentation strategy for the training set and validation set is adopted to avoid overfitting caused by insufficient

Table 1 Chemical cardiotoxicity datasets for the modeling and analysis

	Heart failure		Arrhythmia		Heart block		Hypertension		Heart attack	
	positive	negative	positive	negative	positive	negative	positive	negative	positive	negative
Training set	523	543	959	784	724	453	1081	706	646	402
Validation set	66	69	244	182	178	133	257	184	132	97
total	589	612	1203	966	902	586	1338	890	778	449

samples. For the other four endpoints with larger datasets (ARR, HB, HP, and HA), a 4:1 split ratio was applied to ensure adequate training data while maintaining the statistical reliability of the validation set.

To assess the chemical space consistency between the training and validation sets in the dataset, chemical space scatterplots were created based on molecular weight (MW) and octanol–water partition coefficient (ALogP), and the data distributions of the five heart toxicity endpoints were analyzed. As shown in Fig. 1, the chemical space scatterplot analysis shows high consistency between the training and validation sets, ensuring the reliability of the model evaluation.

Model development results and analysis

For each endpoint, a total of 110 models based on seven machine learning algorithms and five deep learning methods were used. Based on the performance of the models, we separately screened several of the best-performing models and constructed the consensus models. The results of external validation of these consensus models are shown in Table 2 and Fig. 2, and the validation results of the corresponding individual model are given in Table S6–S10.

Among the HF models, four models performed best with ACC values > 0.69, AUC values > 0.77, and stable BA and MCC values. On the basis of these models, a simple averaging method was used to construct a consensus model to reduce the bias of individual models and further improve the prediction performance. The results showed that the AUC of the consensus model increased to 0.86, ACC increased to 0.81, SE and SP increased to 0.79 and 0.83, respectively, showing stronger robustness and reducing the risk of overfitting. In terms of arrhythmia prediction, the external validation results of the consensus model show that the ACC is 0.85 and the AUC is 0.88, which has strong predictive ability and good generalization. Similarly, in the hypertension HP model, the consensus model achieved an AUC of 0.95 and ACC of 0.87 on external validation, clearly outperforming other individual models. For the prediction of HB and HA, the performance of individual models varied significantly. Although some models showed good performance, there was still considerable fluctuation. With the consensus modeling approach, the AUC of the models improved to 0.79 (HB) and 0.82 (HA), and the ACC increased to 0.75 and 0.81, respectively. Particularly in the external validation of HA, the consensus model achieved an MCC of 0.67, demonstrating its reliability in practical applications.

When comparing the models developed in this study with those from previous research, it was found that the consensus models in this study outperformed the single models and traditional QSAR models reported in existing literature across all cardiotoxicity endpoints [2, 9]. The consensus model, by integrating the strengths of multiple individual models, overcame the limitations of traditional models in handling complex cardiotoxicity prediction tasks. It demonstrated stronger stability and accuracy, particularly in predicting different toxicity endpoints.

Molecular property analysis of cardiotoxic and non-cardiotoxic compounds

In this study, we analyzed the differences in the physicochemical properties of cardiotoxic and non-cardiotoxic compounds (as shown in Fig. S1–S5).

The cardiotoxic compounds exhibited higher MW values, suggesting that larger molecules may have a greater propensity to interact with biological targets within the heart, such as ion channels and receptors. For example, HA positive group had a mean MW of 434.56, while the HA negative group averaged 320.45. This observation aligns with previous studies where higher molecular weight often correlates with enhanced toxicity due to increased interaction potential with cellular components [36].

Regarding lipophilicity (logP), cardiotoxic compounds generally showed higher values, especially in the HB and HA positive groups. AlogP values for these compounds were consistently above 2.0, indicating their higher lipid solubility, which facilitates their accumulation in lipid-rich heart tissues, increasing the likelihood of toxicity [37]. These results are consistent with studies that have highlighted the role of lipophilic properties in the enhanced toxicity of drugs, as lipid-soluble compounds tend to penetrate biological membranes more readily [38].

The analysis also revealed significant differences in hydrogen bonding capacity between cardiotoxic and non-cardiotoxic compounds. Compounds with fewer HBD and HBA were more likely to exhibit cardiotoxicity. For instance, the HF positive group showed a mean HBD of 1.97, significantly lower than the non-cardiotoxic group (2.63), suggesting that a reduced hydrogen bonding capability may enhance the membrane permeability of toxic compounds [39].

Moreover, the TPSA was found to be lower in cardiotoxic compounds, especially in the HF and HA positive groups. This indicates that reduced TPSA may enhance the compound's ability to pass through the cell membrane and accumulate in cardiac tissue [40]. These findings support the hypothesis that TPSA can be a useful descriptor in predicting the ability of compounds to penetrate cell membranes and reach their targets in the heart.

The cardiotoxic compounds generally had a lower number RBC. This suggests that more flexible molecules may have an increased ability to adopt favorable conformations for binding to biological targets, such as ion channels and receptors, in the heart. For instance, the HP positive group exhibited a mean RBC of 6.5, higher than that of the

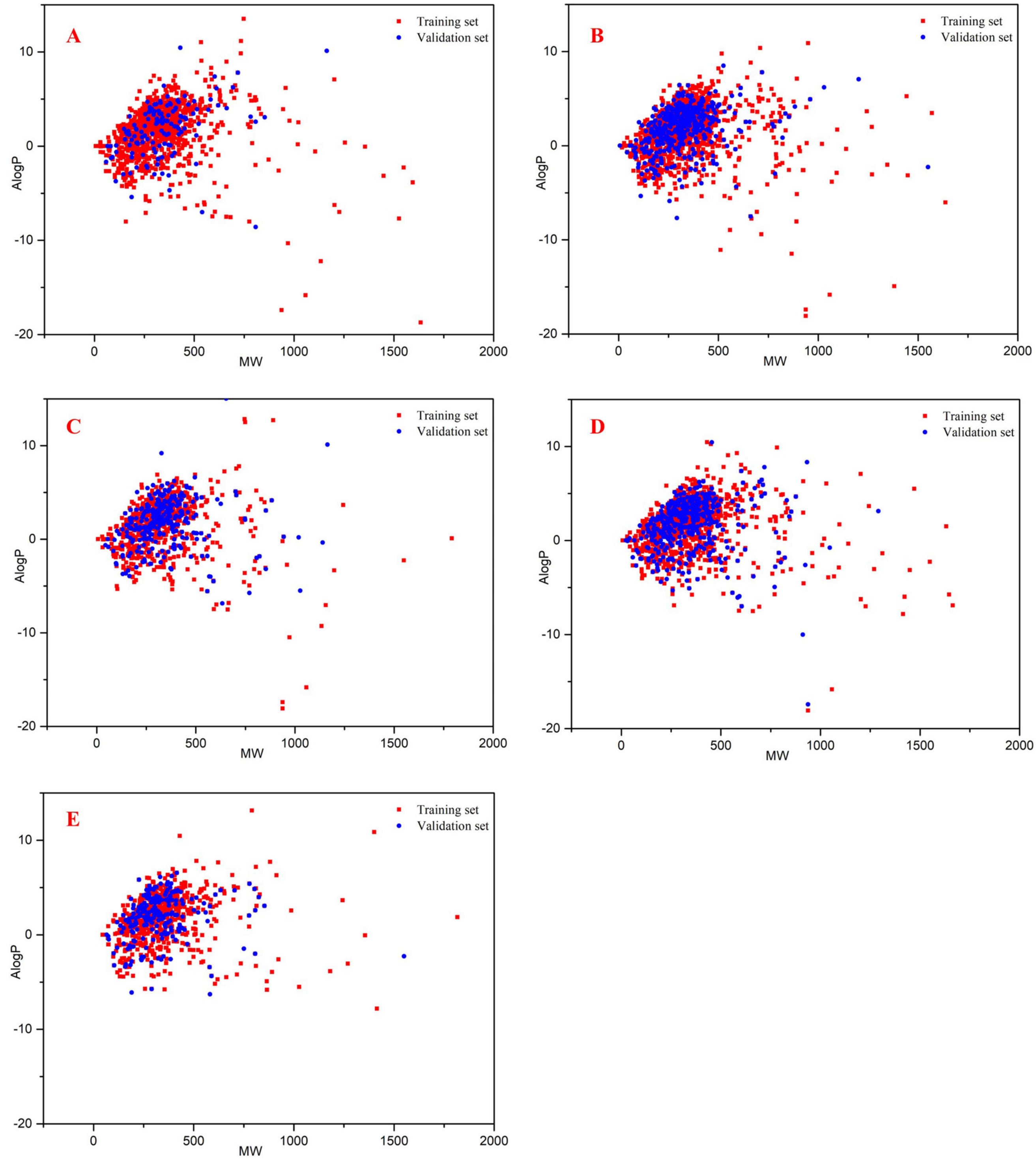

Fig. 1 Chemical space defined by MW and AlogP for the training and validation sets. Red squares indicate the training set, blue circles indicate the validation set. **A** Heart failure datasets; **B** arrhythmia datasets; **C** heart block datasets; **D** hypertension datasets; and **E** heart attack datasets

negative group, which indicates that flexibility in molecular structure might contribute to the selective toxicity of certain compounds [41].

Overall, cardiotoxic compounds have high lipophilicity, molecular size, and hydrogen bonding ability, which is consistent with the following view: These properties promote

Table 2 External validation results of consensus models

Model	ACC	SE	SP	BA	MCC	AUC
Heart failure	0.81	0.79	0.83	0.81	0.61	0.86
Arrhythmia	0.83	0.79	0.87	0.83	0.66	0.89
Heart block	0.75	0.82	0.66	0.74	0.49	0.82
Hypertension	0.87	0.79	0.99	0.89	0.77	0.95
Heart attack	0.78	0.80	0.74	0.77	0.54	0.84

the interaction with targets in cardiac tissue. The research results also emphasized the importance of hydrogen bonds and molecular flexibility in determining the toxicity of compounds. These insights are consistent with previous research reports, which found that hydrophobicity and larger molecular weight are common characteristics of cardiotoxic compounds.

Structural alerts responsible for drug-induced cardiotoxicity

Structural alerts (SAs) can provide important hints for the mechanism of toxic effects and helps to link molecular characteristics with the observed toxicological results. In this study, by analyzing molecular substructures and their occurrence in cardiotoxic and non-cardiotoxic compounds, we identified the key structural alerts typically associated with various cardiotoxic endpoints. Using a frequency-based approach, we identified several key structural alerts strongly associated with drug cardiotoxicity, as shown in Fig. 3. These alerts included secondary amines, benzene ring derivatives, sulfonamide and sulfonylurea groups, and heterocyclic compounds. The presence of these fragments in a chemical is often linked to specific cardiotoxicity mechanisms, such as ion channel blockade, oxidative stress, and disturbances in cellular calcium homeostasis.

Secondary amines, especially those involved in alkylation reactions, were found to be common in compounds associated with HF and ARR. These groups can participate in nucleophilic substitution reactions with biological nucleophiles [42], such as cysteine and lysine residues, leading to irreversible binding with critical proteins involved in cardiac function. This mechanism is known to contribute to the cardiotoxicity of various chemotherapy agents, including doxorubicin. Structurally, secondary amines often increase molecular weight and introduce basicity, while contributing to hydrogen bond donor capacity. These features can enhance both membrane permeability and binding affinity to intracellular cardiac targets, thereby increasing toxic potential.

Halogenated Benzene rings, especially those substituted with electron-withdrawing groups such as $-NO_2$, –Cl, or –F, were frequently observed in compounds linked to HP and HA. The presence of these derivatives can lead to the formation of reactive electrophilic species that damage cardiac tissue through oxidative stress and inflammation [43]. From a physicochemical standpoint, these moieties increase logP, reduce TPSA, and enhance passive diffusion through lipid membranes—characteristics that facilitate tissue accumulation and nonspecific interactions in the heart.

The nitro group, for instance, can undergo reduction reactions, forming reactive nitrogen species (RNS) that contribute to endothelial dysfunction and vascular damage [44]. Sulfonamide and sulfonylurea groups were enriched in compounds associated with cardiac conduction disorders, particularly heart block. These polar functional groups contribute significantly to HBA counts and increase TPSA. Despite their polarity, their presence is often associated with interference in ion channel regulation, possibly via interaction with sodium or potassium regulatory enzymes [45]. Notably, the interplay between increased polarity and enzyme binding underscores the complexity of structure–toxicity relationships. Nitrogen-containing heterocycles, such as triazoles and pyridines, were commonly found in cardiotoxic compounds, particularly those causing arrhythmia and myocardial infarction (MI). These ring systems often interact with cardiac ion channels, especially the hERG potassium channel, contributing to QT interval prolongation and life-threatening arrhythmias [46]. Their relatively small size and aromaticity enable strong π–π and hydrogen bonding interactions with channel proteins, while maintaining moderate logP and TPSA values conducive to membrane permeability.

The identified SAs collectively shape the physicochemical property landscape of cardiotoxic compounds. Specifically, cardiotoxic molecules tend to exhibit elevated MW and logP, along with moderate-to-low TPSA and reduced HBD/HBA counts, all of which enhance tissue permeability and target interaction in cardiac environments. These findings offer valuable guidance for rational drug design, enabling early-stage elimination or modification of hazardous structural motifs.

Availability of data and models

The data used in this study were obtained from publicly available databases and underwent standardized data curation and preprocessing. All datasets are provided in the Supplementary Materials.

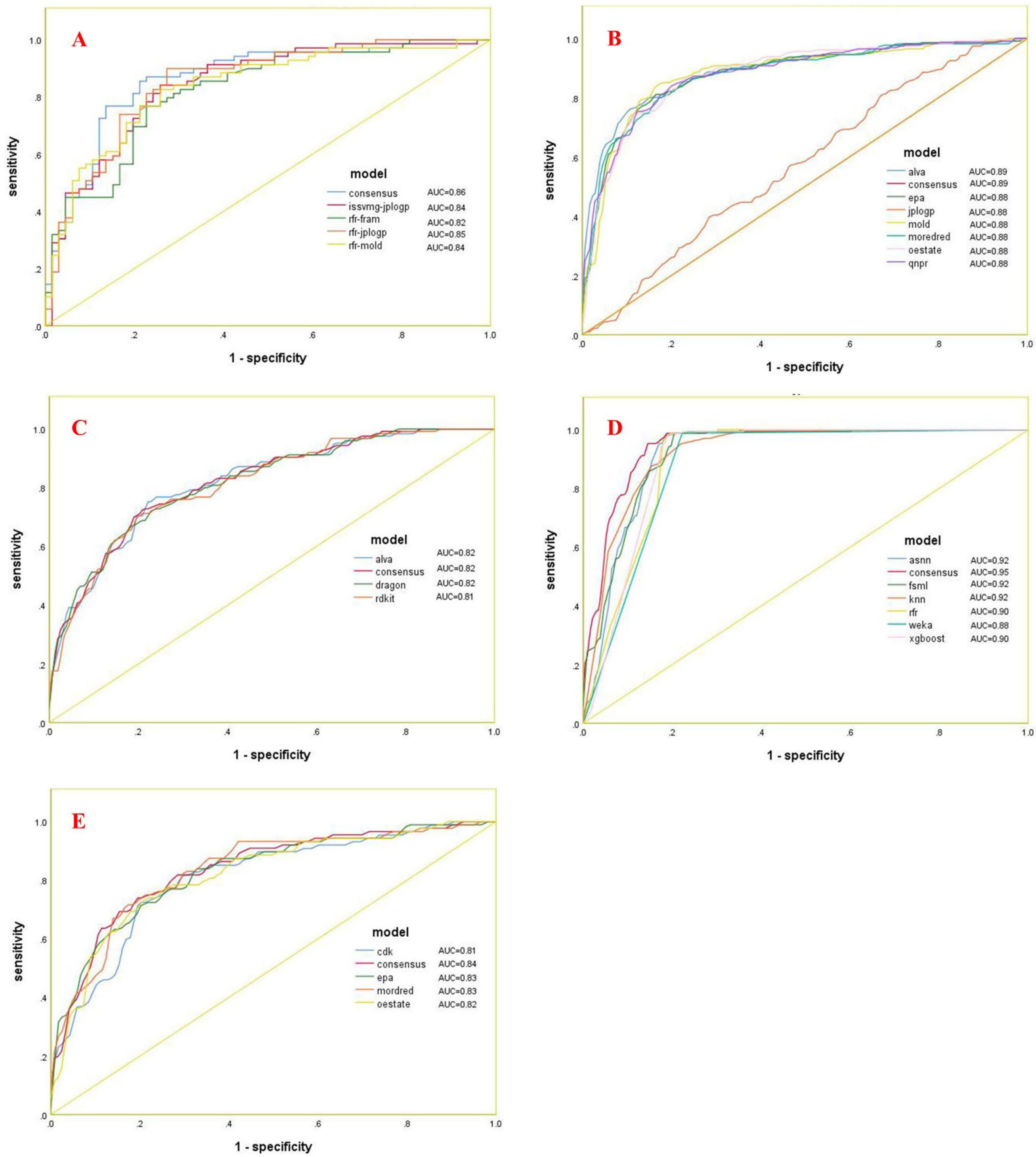

Fig. 2 ROC curves of the models for external validation. Each colored line represents a different model. **A** Heart failure datasets; **B** arrhythmia datasets; **C** heart block datasets; **D** hypertension datasets; and **E** heart attack datasets

The consensus models developed in this study can be accessed via https://ochem.eu/article/166881. The corresponding individual machine learning models can also be accessed through the consensus models. Additionally, the chemical structures and descriptors used for training and validation can be downloaded from the model pages, allowing researchers to replicate or further optimize the models.

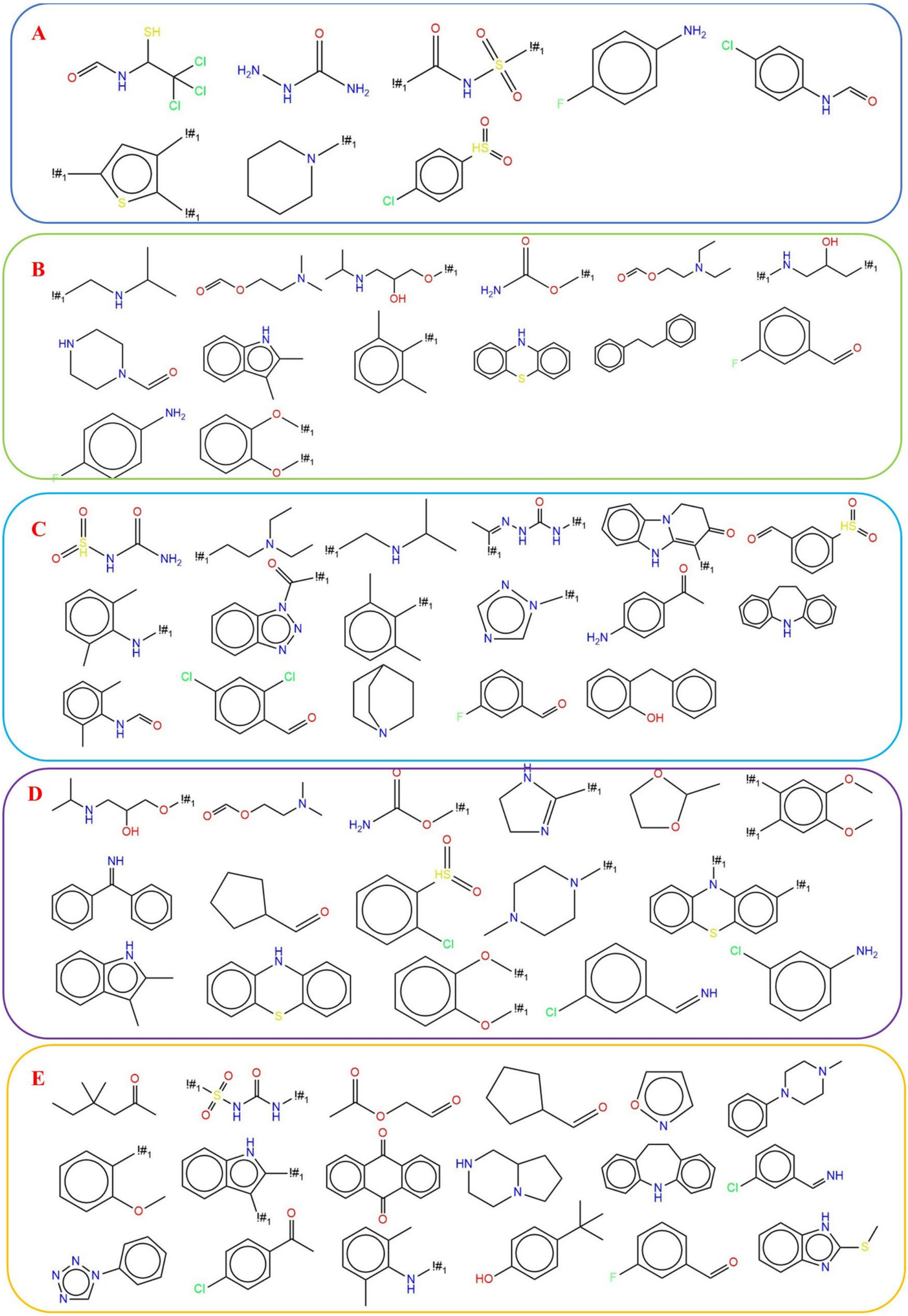

Fig. 3 Representative structural alerts responsible for chemical cardiotoxicity. **A** Heart failure chemicals; **B** arrhythmia chemicals; **C** heart block chemicals; **D** hypertension chemicals; and **E** heart attack chemicals

Conclusion

In this study, a series of machine learning and deep learning models were developed for the prediction of drug-induced cardiotoxicity across five key endpoints, including heart failure, arrhythmia, heart block, hypertension, and heart attack. Based on the best-performing individual classifiers, consensus models were developed, consistently outperforming individual models in cross-validation and external validation. Further analysis of molecular properties revealed that higher molecular weight, increased lipophilicity, lower hydrogen bonding capacity, and reduced topological polar surface area were significantly associated with cardiotoxic compounds. These findings suggest that compound permeability, bioavailability, and ion channel interactions may contribute to drug-induced cardiotoxicity. Additionally, structural alerts strongly associated with cardiotoxicity were identified, including secondary amines, benzene derivatives, sulfonamide/sulfonylurea groups, and heterocyclic structures. Mechanistic analysis indicated that these SAs may contribute to ion channel inhibition, oxidative stress induction, and calcium homeostasis disruption, providing valuable insights into the molecular basis of cardiotoxicity. This study offers a reliable approach for assessing cardiotoxicity and contribute to a deeper understanding of the structural determinants of drug-induced cardiotoxicity from a molecular perspective.

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Author contributions S.L, H.X, F.L, and R.N contributed in the form of conceptualization and drafting of the manuscript. S.L, H. X, Y. S, and X.L contributed in the form of redefining conceptualization, figures. All authors reviewed the manuscript.

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Data availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

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